

Patch Compliance Report Using Ansible

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Exclusive to: Dabin Distro

Purpose

This project automates **patch compliance reporting** across multiple Linux nodes using **Ansible**. It eliminates the need for agents — relying solely on **passwordless SSH** and **sudo privileges** for the ubuntu user. The controller node collects patch data, generates compliance metrics, and emails a detailed report with attachments.

Configuration Overview

Component	Description
SSH Setup	Passwordless SSH from controller to all nodes using ubuntu user
Sudo Privilege	Passwordless sudo configured for ubuntu user on all nodes
Software Requirements	ansible-core on controller, python3 on all nodes
Controller Files	ansible.cfg and inventory configured
Mail Credentials	Stored securely in cred.yml (Ansible Vault)

File Structure

Code

```
├── ansible.cfg
├── applied_patches.txt.j2
├── cred.yml
├── inventory
├── mail.html.j2
├── pending_patches.txt.j2
└── report.yml
```

Workflow Summary

Play 1 – Node Operations

- Runs on all nodes (including controller).
- Updates apt caches.
- Collects patch data and sets custom facts:
 - Pending patch count & list
 - Applied patch count & list

Play 2 – Controller Operations

- Runs only on controller.
- Renders templates:
 - mail.html.j2 → HTML mail body
 - applied_patches.txt.j2 → Applied patches list
 - pending_patches.txt.j2 → Pending patches list
- Generates files:
 - /tmp/applied_patches.txt
 - /tmp/pending_patches.txt
- Sends email with:
 - HTML compliance table
 - Two text attachments

Compliance Formula

$$Compliance = \frac{AppliedPatchCount}{AppliedPatchCount + PendingPatchCount} \times 100$$



Pending Patches Summary

A screenshot of a Gmail interface on a dark-themed desktop. The Gmail sidebar on the left shows standard navigation options like 'Compose', 'Inbox', 'Starred', 'Snippets', 'Sent', 'Drafts', 'Purchases', and 'More'. Below these are 'Labels' such as 'Aria', 'CoinDCX', 'Flipkart', 'Gleadow', 'Idle', 'Jin', 'LinkedIn', 'MakeMyTrip', 'Microsoft', 'now', 'Star Health', and 'Zepto'. The main area of the Gmail window displays a terminal window with a white background and black text. The terminal output shows system update logs for four different nodes: 'controller', 'node1', 'node2', and 'node4'. Each node's log starts with 'Patches Pending: [number]' followed by a 'Listing...' command. The logs list various software packages and their versions, along with their upgrade status. For example, 'controller' shows updates for 'power-profiles-daemon' and 'python3-software-properties'. 'node1' shows updates for 'python3-software-properties/resolute-updates' and 'software-properties-common'. 'node2' shows no pending updates. 'node3' shows updates for 'linux-aws/resolute-updates', 'linux-headers-aws/resolute-updates', and 'linux-image-aws/resolute-updates'. 'node4' shows no pending updates. The terminal window is titled 'terminal' in the top right corner.

Applied Patches Log

applied_patches.txt

Open with Google Docs

controller

Date	Time	Action	Package Name
2026-09-06	11:02:27	upgrade	bsdutils:amd64 1.2.39.3-9ubuntu6.5 1.2.39.3-9ubuntu6.6
2026-09-06	11:02:28	upgrade	coreutils:amd64 9.4-3ubuntu6.2 9.4-3ubuntu6.3
2026-09-06	11:02:29	upgrade	diffutils:amd64 1.3.10-1build1 1.3.10-1ubuntu1
2026-09-06	11:02:29	upgrade	ncurses-bin:amd64 6.4-20240113-1ubuntu2 1 6.4-20240113-1ubuntu2.2
2026-09-06	11:02:30	upgrade	libperl5.38t64:amd64 5.38.2-3.2ubuntu0.3 5.38.2-3.2ubuntu0.4
2026-09-06	11:02:30	upgrade	perl:amd64 5.38.2-3.2ubuntu0.3 5.38.2-3.2ubuntu0.4
2026-09-06	11:02:30	upgrade	perl-base:amd64 5.38.2-3.2ubuntu0.3 5.38.2-3.2ubuntu0.4
2026-09-06	11:02:31	upgrade	perl-modules-5.38:all 5.38.2-3.2ubuntu0.3 5.38.2-3.2ubuntu0.4
2026-09-06	11:02:32	upgrade	hzip2:amd64 1.0.8-5.1build0 1.0.8-5.1ubuntu1
2026-09-06	11:02:32	upgrade	libbz2-1.0:amd64 1.0.8-5.1build0 1.0.8-5.1ubuntu1
2026-09-06	11:02:32	upgrade	zlib1g:amd64 1.1.3.dfsg-3.1ubuntu2 1.1.3.dfsg-3.1ubuntu2.2
2026-09-06	11:02:33	upgrade	util-linux:amd64 2.39.3-9ubuntu6.5 2.39.3-9ubuntu6.6
2026-09-06	11:02:35	upgrade	ncurses-base:all 6.4-20240113-1ubuntu2 1 6.4-20240113-1ubuntu2.2
2026-09-06	11:02:36	upgrade	ncurses-term:all 6.4-20240113-1ubuntu2 1 6.4-20240113-1ubuntu2.2
2026-09-06	11:02:37	upgrade	libpam0g:amd64 1.5.3-5ubuntu5.6 1.5.3-5ubuntu5.7
2026-09-06	11:02:37	upgrade	systemd-timesyncd:amd64 255.4-1ubuntu8.16 255.4-1ubuntu8.17
2026-09-06	11:02:37	upgrade	libss13t64:amd64 3.0.13-0ubuntu3.12 3.0.13-0ubuntu3.15
2026-09-06	11:02:38	upgrade	systemd-resolved:amd64 255.4-1ubuntu8.16 255.4-1ubuntu8.17
2026-09-06	11:02:38	upgrade	systemd-oemd:amd64 255.4-1ubuntu8.16 255.4-1ubuntu8.17
2026-09-06	11:02:38	upgrade	libblkid1:amd64 2.39.3-9ubuntu6.5 2.39.3-9ubuntu6.6
2026-09-06	11:02:39	upgrade	libcrypt20:amd64 1.10.3-2ubuntu0.1 1.10.3-2ubuntu0.2
2026-09-06	11:02:39	upgrade	libmount1:amd64 2.39.3-9ubuntu6.5 2.39.3-9ubuntu6.6
2026-09-06	11:02:40	upgrade	libos-proton:amd64 255.4-1ubuntu8.16 255.4-1ubuntu8.17
2026-09-06	11:02:40	upgrade	systemd-dev:all 255.4-1ubuntu8.16 255.4-1ubuntu8.17
2026-09-06	11:02:40	upgrade	systemd:amd64 255.4-1ubuntu8.16 255.4-1ubuntu8.17
2026-09-06	11:02:40	upgrade	udev:amd64 255.4-1ubuntu8.16 255.4-1ubuntu8.17
2026-09-06	11:02:41	upgrade	libsystemd-shared:amd64 255.4-1ubuntu8.16 255.4-1ubuntu8.17
2026-09-06	11:02:41	upgrade	libsystemd:amd64 255.4-1ubuntu8.16 255.4-1ubuntu8.17
2026-09-06	11:02:43	upgrade	systemd-sysv:amd64 255.4-1ubuntu8.16 255.4-1ubuntu8.17
2026-09-06	11:02:43	upgrade	libpam-systemd:amd64 255.4-1ubuntu8.16 255.4-1ubuntu8.17
2026-09-06	11:02:43	upgrade	libudev1:amd64 255.4-1ubuntu8.16 255.4-1ubuntu8.17
2026-09-06	11:02:44	upgrade	libnss-runtime:all 1.5.3-5ubuntu5.6 1.5.3-5ubuntu5.7
2026-09-06	11:02:44	upgrade	libuid1:amd64 2.39.3-9ubuntu6.5 2.39.3-9ubuntu6.6
2026-09-06	11:02:45	upgrade	libfdisk1:amd64 2.39.3-9ubuntu6.5 2.39.3-9ubuntu6.6
2026-09-06	11:02:45	upgrade	mount:amd64 2.39.3-9ubuntu6.5 2.39.3-9ubuntu6.6
2026-09-06	11:02:45	upgrade	libpam-modules-bin:amd64 1.5.3-5ubuntu5.6 1.5.3-5ubuntu5.7
2026-09-06	11:02:46	upgrade	libpam-modules:amd64 1.5.3-5ubuntu5.6 1.5.3-5ubuntu5.7
2026-09-06	11:02:47	upgrade	console-setup-linux:all 1.226ubuntu1 1.226ubuntu1.1
2026-09-06	11:02:47	upgrade	console-setup:all 1.226ubuntu1 1.226ubuntu1.1
2026-09-06	11:02:47	upgrade	keyboard-configuration:all 1.226ubuntu1 1.226ubuntu1.1
2026-09-06	11:02:48	upgrade	python3.12-venv:amd64 3.12.3-1ubuntu0.15 3.12.3-1ubuntu0.16
2026-09-06	11:02:48	upgrade	libpython3.12t64:amd64 3.12.3-1ubuntu0.15 3.12.3-1ubuntu0.16
2026-09-06	11:02:48	upgrade	python3.12:amd64 3.12.3-1ubuntu0.15 3.12.3-1ubuntu0.16
2026-09-06	11:02:48	upgrade	libpython3.12-stdlib:amd64 3.12.3-1ubuntu0.15 3.12.3-1ubuntu0.16
2026-09-06	11:02:49	upgrade	python3.12-minimal:amd64 3.12.3-1ubuntu0.15 3.12.3-1ubuntu0.16
2026-09-06	11:02:49	upgrade	libpython3.12-minimal:amd64 3.12.3-1ubuntu0.15 3.12.3-1ubuntu0.16
2026-09-06	11:02:49	upgrade	libncurses6:amd64 6.4-20240113-1ubuntu2 1 6.4-20240113-1ubuntu2.2
2026-09-06	11:02:49	upgrade	libncursesw6:amd64 6.4-20240113-1ubuntu2 1 6.4-20240113-1ubuntu2.2
2026-09-06	11:02:49	upgrade	libtinfo6:amd64 6.4-20240113-1ubuntu2 1 6.4-20240113-1ubuntu2.2
2026-09-06	11:02:50	upgrade	libsmartcols1:amd64 2.39.3-9ubuntu6.5 2.39.3-9ubuntu6.6

Email Report Example

Patch Complian Report | 13/09/2026-19:35:56

Inbox x



linuxadminari@gmail.com

to me ▾

7:36 PM (21 minutes ago)



Hi Team,

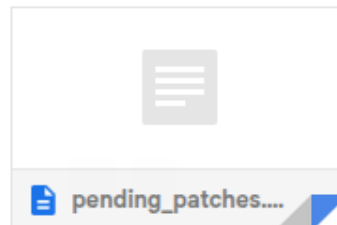
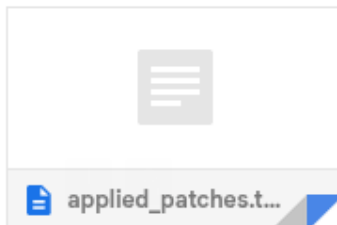
Kindly refer below compliance report and take appropriate action.
Applied and pending Patches list is attached for reference.

Hostname	Ip Address	Patch Pending	Patch Applied	Patch Compliance
controller	192.168.29.95	3	212	98 %
node1	172.31.44.86	5	159	96 %
node2	172.31.38.222	0	159	100 %
node3	172.31.32.29	17	156	90 %
node4	172.31.44.168	0	159	100 %

Regards,

Admin Team

2 Attachments • Scanned by Gmail ⓘ ⬇️ 🗑️ Add all to Drive



✓ Key Takeaways

- **Agentless automation** using SSH and sudo.
- **Centralized compliance reporting** with HTML and TXT outputs.
- **Secure credential management** via Ansible Vault.
- **Scalable design** — easily extendable to more nodes.